ALLSAFE Security Analysis Report: brute_force_windows.csv

- Analysis ID: ds-6e8498492f01
- Report ID: RPT-7B3CE04101
- Generated: 2026-10-31T04:31:17.891817+00:00
- Log file: brute_force_windows.csv
- Log type: Windows authentication events
- Analysis started: 2026-10-31T04:31:16.678600+00:00
- Analysis completed: 2026-10-31T04:31:17.754089+00:00

Overall security assessment

THREATS DETECTED

2 detection(s) have strong supporting evidence and need investigation.

Executive summary

brute_force_windows.csv: 15 of 15 records were processed, producing 2 alert(s) and 1 incident(s). The highest priority is INC-000005, Successful Login After Repeated Failures (dc-01.corp.example), at 74/100 (HIGH). 4 recommended action(s) are waiting for an analyst. No action has changed a real system.

Threat overview

- Attack categories: Credential Access, Initial Access
- Rules fired: IOA-001, IOA-002

Processing summary

- Status: COMPLETED
- Total uploaded records: 15
- Successfully processed records: 15
- Rejected records: 0
- Duplicate records: 0

Data quality

- Malformed records: 0
- Invalid timestamps: 0
- Missing timestamps: 0
- Flows with no timestamp, kept: 0
- Missing optional values: dst_ip 15
- Fields with no equivalent in the common schema: TimeCreated, EventID, TargetUserName, LogonType, Channel
- Warning: 5 field(s) have no equivalent in the common schema. They are preserved with each event, but no detection reads them.

Detection engines

Executed

- IOC enrichment: evaluated 15 of 15 event(s), 0 finding(s)
- IOA behavior rules: evaluated 15 of 15 event(s), 2 finding(s)
- UEBA baselines: evaluated 6 of 15 event(s), 0 finding(s)
- Anomaly model: evaluated 15 of 15 event(s), 1 finding(s)

Skipped or failed

- Supervised ML (saved network model) (SKIPPED): This data has no records with all 78 CIC-IDS2017 flow features the model needs. Logs of other kinds are never fed to it.

Finding classification

- Confirmed detections: 2 alert(s) over 10 event(s), 1 incident(s)
- Suspicious activity: 0 alert(s) over 0 event(s), 1 anomaly window(s), 0 flow(s) classified as attacks
- Benign activity (normal events): 5 event(s). These events triggered no applicable detection condition. That is not proof they are safe.
- Data quality issues: 0 bad record(s), 0 duplicate(s)
- Detection limitations: 1
  - Supervised ML (saved network model): This data has no records with all 78 CIC-IDS2017 flow features the model needs. Logs of other kinds are never fed to it.

Detection results

- Alerts: 2
- Incidents: 1

IOC matches (0)

- 15 event(s) were checked against the threat intelligence database.
- No known indicator was matched. This does not mean these events are safe: the feed only holds known indicators.

IOA rule matches (2)

- IOA-002 Successful Login After Repeated Failures (Confirmed detection)
  - Severity HIGH, confidence 89%, risk 55/100 (MEDIUM), 10 event(s), last seen 2026-01-15T09:01:30+00:00
  - Description: A login succeeded from a source that had just failed repeatedly, which suggests the password was guessed.
  - Why it fired: 203.0.113.44 logged in successfully after 9 failed attempts within 600s.
  - Triggering conditions: {"type": "sequence", "group_by": ["src_ip"], "threshold": 5, "window_seconds": 600, "first": {"field": "event_type", "equals": "authentication_failure"}, "then": {"field": "event_type", "equals": "authentication_success"}}
  - MITRE ATT&CK: T1110 Brute Force, T1078 Valid Accounts
  - Supporting event IDs: ds-6e8498492f01-d1fd1da35204e42f, ds-6e8498492f01-810fe33728e23854, ds-6e8498492f01-2efcab1eb1c79387, ds-6e8498492f01-1015ae535cf876d9, ds-6e8498492f01-5ca72c1fe136f8e1, ds-6e8498492f01-85331f5ad1684eae, ds-6e8498492f01-f5bc3c613ab89a60, ds-6e8498492f01-316bb973e12fc286, ds-6e8498492f01-af1a7d29c4b2160d, ds-6e8498492f01-0303cb6d4831d065
  - Evidence: Failed login for svc_backup from 203.0.113.44 to dc-01.corp.example
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  - Evidence: Failed login for svc_backup from 203.0.113.44 to dc-01.corp.example
- IOA-001 Brute Force Login Attempts (Confirmed detection)
  - Severity MEDIUM, confidence 74%, risk 41/100 (MEDIUM), 9 event(s), last seen 2026-01-15T09:01:04+00:00
  - Description: Many failed logins from one source address in a short time.

  - Why it fired: 9 failed logins from 203.0.113.44 within 300s (threshold 5).
  - Triggering conditions: {"type": "threshold", "group_by": ["src_ip"], "threshold": 5, "window_seconds": 300, "match": {"field": "event_type", "equals": "authentication_failure"}}
  - MITRE ATT&CK: T1110 Brute Force
  - Supporting event IDs: ds-6e8498492f01-d1fd1da35204e42f, ds-6e8498492f01-810fe33728e23854, ds-6e8498492f01-2efcab1eb1c79387, ds-6e8498492f01-1015ae535cf876d9, ds-6e8498492f01-5ca72c1fe136f8e1, ds-6e8498492f01-85331f5ad1684eae, ds-6e8498492f01-f5bc3c613ab89a60, ds-6e8498492f01-316bb973e12fc286, ds-6e8498492f01-af1a7d29c4b2160d
  - Evidence: Failed login for svc_backup from 203.0.113.44 to dc-01.corp.example
  - Evidence: Failed login for svc_backup from 203.0.113.44 to dc-01.corp.example
  - Evidence: Failed login for svc_backup from 203.0.113.44 to dc-01.corp.example

UEBA findings (0)

- None.

ML classification results

- The saved network model classified nothing in this data. See the detection engines above for why.

Anomaly detection results

- Unusual activity windows: 1. Statistically unusual activity, not confirmed attacks.
- 203.0.113.44 (strength 73 of 100, 10 event(s)): Unusual compared with the model's synthetic baseline of normal activity: 9 failed logins (typically about 0.0), 10 events (typically about 1.6), 1 successful logins (typically about 0.3). A statistical anomaly, not a confirmed attack.

Incident investigation

INC-000005: Successful Login After Repeated Failures (dc-01.corp.example)

- Status: ACTIVE, priority P2
- Severity: HIGH
- Confidence: 97%
- Risk: 74/100 HIGH
- Affected hosts: dc-01.corp.example
- Affected users: svc_backup
- Related alerts: IOA-002, IOA-001
- Possible ATT&CK stages: T1078 Valid Accounts (Initial Access); T1110 Brute Force (Credential Access)
- Automation: Needs SOC approval: every response action waits for an analyst decision. Evidence is collected and an urgent notification is sent.

Why this score:

- severity (18.8 points): Incident severity is HIGH.
- evidence confidence (24.2 points): Combined evidence confidence is 97%.
- ioc ioa signals (16.0 points): 2 distinct behavior rule(s), 0 threat intelligence match(es).
- behavioral anomalies (10.9 points): 1 statistically unusual activity window(s), strength 73 of 100. The strongest one counts.
- attack progression (4.4 points): 2 MITRE tactic(s) across 2 distinct detection(s):

Credential Access, Initial Access.

Timeline:

- 2026-01-15T09:00:00+00:00: Failed login for svc_backup from 203.0.113.44 to dc-01.corp.example (x9) [IOA-001, IOA-002]
- 2026-01-15T09:01:30+00:00: Successful login for svc_backup from 203.0.113.44 to dc-01.corp.example [IOA-002]

Affected users and hosts

- Users: svc_backup
- Hosts: dc-01.corp.example

Risk analysis

- Severity distribution (alerts): HIGH 1, MEDIUM 1
- Incident risk distribution: CRITICAL 0, HIGH 1, MEDIUM 0, LOW 0
- Alert risk distribution: CRITICAL 0, HIGH 0, MEDIUM 2, LOW 0
- Risk is an investigation priority, not the probability that an attack is real. Each incident above lists the factors behind its score.

MITRE ATT&CK mappings

- T1078 Valid Accounts (Initial Access)
- T1110 Brute Force (Credential Access)

Recommended remediation

Historical analysis is read-only. Nothing below has been done, and any disruptive step needs an analyst to approve it.

- IOA-001 Brute Force Login Attempts: Review the targeted accounts and whether this source address is expected. Consider rate limiting or blocking the source address after review.
- IOA-002 Successful Login After Repeated Failures: Confirm with the account owner whether this login was theirs. If not, reset the account's password and revoke its active sessions. Review what the account did after this login.

Recommended actions (simulation only)

- INC-000005, PENDING: Investigate source IP on 203.0.113.44
- INC-000005, PENDING: Review affected account on svc_backup
- INC-000005, PENDING: Block test IP on 203.0.113.44 (disruptive, needs approval)
- INC-000005, PENDING: Disable test account on svc_backup (disruptive, needs approval)

Analyst decisions

- None yet.

Model performance

- This data has no ground truth labels, so these are descriptive counts, not performance metrics.

Known limitations

- Detection combines behavior rules, threat intelligence matches, statistical baselines and two models. None of them proves an attack happened, and none proves that one did not.
- The saved network model runs only on flow data with its 78 CIC-IDS2017 features. It has not been validated here, no preprocessing pipeline was saved with it, and its output is uncalibrated, so no accuracy or prediction confidence is claimed.
- The anomaly model was trained on synthetic normal activity, and does not run on live telemetry. Its findings are statistical anomalies, not confirmed attacks.
- Threat intelligence only covers known indicators, and the bundled feed is synthetic. No match does not mean an event is safe.
- Performance metrics compare results with labels on synthetic or supplied data. They do not predict performance on real attacks.
- The synthetic scenarios and the detection rules were written by the same team, so scenario results show the pipeline works as designed.
- Behavioral checks need history. An entity with too little history is not judged.
- Window-based rules cite every matching event in their window, so a normal event inside an attack window counts as an event-level false positive.
- Risk is an investigation priority, not the probability that an attack is real.
- Historical analysis is read-only. Every response action is a recommendation that needs an analyst, and runs only in simulation. No firewall, account, host or process is changed.
- Analyst names are self-reported: there are no user accounts.